

ShellSort Algorithm

4. $[5, 10, 60, 0, -1, 34, 6, 10]$
_{0 1 2 3 4 5 6 7}

$$\text{Gap} = 8/2 = 4$$

$[5, 10, 60, 0, -1, 34, 6, 10]$

$[-1, 10, 60, 0, 5, 34, 6, 10]$

← No swap

$[-1, 10, 60, 0, 5, 34, 6, 10]$

$[-1, 10, 6, 0, 5, 34, 60, 10]$

← No swap

$[-1, 10, 6, 0, 5, 34, 60, 10]$

← No swap

$$\text{Gap} = 8/4 = 2$$

$[-1, 10, 6, 0, 5, 34, 60, 10]$

$[-1, 0, 6, 10, 5, 34, 60, 10]$

$[-1, 0, 5, 10, 6, 34, 60, 10]$

← No swap

$[-1, 0, 5, 10, 6, 34, 60, 10]$

← No swap

$[-1, 0, 5, 10, 6, 34, 60, 10]$

$[-1, 0, 5, 10, 6, 10, 60, 34]$

$$\text{Gap} = 8/8 = 1 \text{ (Insertion Sort)}$$

$[-1, 0, 5, 10, 6, 10, 60, 34]$

$[-1, 0, 5, 10, 6, 10, 60, 34]$

$[-1, 0, 5, 10, 6, 10, 60, 34]$

$[-1, 0, 5, 6, 10, 10, 60, 34]$

$[-1, 0, 5, 6, 10, 10, 60, 34]$

$[-1, 0, 5, 6, 10, 10, 60, 34]$

result[] = $[-1, 0, 5, 6, 10, 10, 34, 60]$